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1. Definition of the zero input response

The input signal is zero, and the response produced by acting solely on the initial state of the system is called the zero input response, which is denoted $y_{\mathrm{zi}}[k]$

Mathematical model:

$$\sum_{i=0}^{n} a_i y[k-i] = 0$$

2. Solving for zero input responses

Solution process:

Step 1: Find the eigenvalue of the difference equation；

Step 2: Determine the form of the zero-input response based on the feature root；

Step 3: Substitute the initial state into the zero input response representation, and solve the pending coefficient to get the zero input response.

2. Solving for zero input responses

Zero input response in the form of $y_{zi}[k]$:

(1) Feature roots appear unequal solid roots r_1, r_2, …, r_n

$$y_{zi}[k] = C_1 r_1^k + C_2 r_2^k + \cdots + C_n r_n^k$$

(2) Feature roots appear equal solid roots $r_1 = r_2 = \ldots = r_n$

$$y_{zi}[k] = C_1 r^k + C_2 k r^k + \cdots + C_n k^{n-1} r^k$$

(3) Characteristic roots appear in pairs of conjugate compound roots $r_{1,2} = a \pm \mathrm{j}b = \rho \mathrm{e}^{\pm \mathrm{j}\Omega_0}$

$$y_{zi}[k] = C_1 \rho^k \cos \Omega_0 k + C_2 \rho^k \sin \Omega_0 k$$

2. Solving for zero input responses

[Ex.] The difference equation for a discrete LTI system is

$y[k]+3y[k-1]+2y[k-2]=x[k], k\geq 0$， The initial state is $y[-1]=0$， $y[-2]= 1/2$，

Find a system zero input response $y_{zi}[k]$

Solution: The characteristic equation is $r^2+3r+2=0$

The feature root is $r_1=-1, r_2=-2$ (Two unequal solid roots)

$$y_{zi}[k]=C_1(-1)^k+C_2(-2)^k$$

$$y[-1]=-C_1-\frac{1}{2}C_2=0$$

$$y[-2]=C_1+\frac{1}{4}C_2=\frac{1}{2}$$

$\Rightarrow$ $C_1=1$， C_2-2

$$y_{zi}[k]=(-1)^k-2(-2)^k \quad k\geq 0$$

2. Solving for zero input responses

[Ex.] The difference equation for a discrete LTI system is

$y[k]+4y[k-1]+4y[k-2]=x[k]$, $k\geq 0$, the initial state is $y[-1]=0$, $y[-2]=1/2$,

Find a system zero input response $y_{zi}[k]$

Solution: The characteristic equation is $r^2+4r+4=0$

The feature root is $r_1=r_2=-2$ (Two equal solid roots)

$$y_{zi}[k]=C_1k(-2)^k+C_2(-2)^k$$

$$y[-1]=\frac{C_1}{2}-\frac{C_2}{2}=0$$

$$y[-2]=-\frac{C_1}{2}+\frac{C_2}{4}=\frac{1}{2}$$

$\Rightarrow$ $C_1=C_2=-2$

$$y_{zi}[k]=-2k(-2)^k-2(-2)^k,\quad k\geq 0$$

2. Solving for zero input responses

[Ex.] The difference equation for a discrete LTI system is

$y[k]+2y[k-1]+2y[k-2]=x[k]$, $k\geq 0$, the initial state is $y[-1]=0$, $y[-2]=1/2$,

Find a system zero input response $y_{zi}[k]$

Solution: The characteristic equation is $r^2+2r+2=0$

The feature root is $r_1=-1+\mathrm{j},\ r_2=-1-\mathrm{j}$

$$r_{1,2}=\sqrt{2}\mathrm{e}^{\pm \mathrm{j}\frac{3\pi}{4}}$$

$$y_{zi}[k]=\sqrt{2}^k\left(C_1\cos\frac{3\pi}{4}k+C_2\sin\frac{3\pi}{4}k\right)$$

$$y[-1]=-C_1-C_2=0$$

$$y[-2]=C_2/2=1/2$$

$$\Rightarrow \quad C_1=-1,\ C_2=1$$

$$y_{zi}[k]=\sqrt{2}^k\left[-\cos\frac{3\pi}{4}k+\sin\frac{3\pi}{4}k\right],\quad k\geq 0$$

Unit impulse response of discrete LTI systems

- Definition of unit impulse response
- Solution of the unit impulse response

1. Definition of unit impulse response

Under conditions where the initial state of the system is zero， The output response produced by the δ[k] excitation system in a unit pulse train， The unit impulse response called the system， Denoted by the symbol $h[k]$.。

（The initial system state is zero）

1. Definition of unit impulse response

If the linear constant coefficient difference equation

for describing a discrete LTI system is

$$\sum_{i=0}^{n} a_i y[k-i] = \sum_{j=0}^{m} b_j x[k-j]$$

Then the unit impulse response h[k] of the discrete LTI system should be satisfied

$$\sum_{i=0}^{n} a_i h[k-i] = \sum_{j=0}^{m} b_j \delta[k-j]$$

2. Solution of the unit impulse response

[Ex.] The difference equation for a causal discrete LTI system is

$$y[k]+3y[k-1]+2y[k-2]=x[k]$$

Find the unit impulse response of the system $h[k]$。

analyse: When the system input signal $x[k]$ is $\delta[k]$ and the output signal $y[k]$ is $h[k]$

The differential equation that describes the system is

$$h[k]+3h[k-1]+2h[k-2]=\delta[k]$$

When $k > 0$, $\delta[k]=0$, the difference equation describing the system is $h[k]+3h[k-1]+2h[k-2]=0$

Therefore, the unit impulse response of the system has a homogeneous solution form

2. Solution of the unit impulse response

[Ex.] The difference equation for a causal discrete LTI system is

$$y[k]+3y[k-1]+2y[k-2]=x[k]$$

Find the unit impulse response of the system $h[k]$

analyse: How to determine the pending coefficient?

The initial conditions of the system are determined by the initial state of the system ($h[0]$、$h[1]$).

It is recursively derived from the difference equation $h[-1] = h[-2] = \ldots = h[-n] = 0$.

This method is called the equivalent initial condition method

2. Solution of the unit impulse response

[Ex.] The difference equation for a causal discrete LTI system is

$$y[k]+3y[k-1]+2y[k-2]=x[k]$$

Find the unit impulse response of the system $h[k]$。

solution: $h[k]$ satisfies the equation $h[k]+3h[k-1]+2h[k-2]=\delta[k]$

(1) Determine the form of $h[k]$.

The characteristic equation is $r^2+3r+2=0$

The feature root is $r_1=-1, r_2=-2$

$$\Downarrow$$

$$h[k]=C_1(-1)^k+C_2(-2)^k, k\geq 0$$

2. Solution of the unit impulse response

[Ex.] The difference equation for a causal discrete LTI system is

$$h[k]+3h[k-1]+2h[k-2]=\delta[k]$$

Find the unit impulse response of the system $h[k]$。

Solution: $h[k]$ satisfies the equation $y[k]+3y[k-1]+2y[k-2]=x[k]$

(2) Find equivalent initial conditions

For causal systems there is $h[-1]=h[-2]=0$, Substituting the above equation can be derived

$$h[0]=\delta[0]-3h[-1]-2h[-2]=1$$
$$h[1]=\delta[1]-3h[0]-2h[-1]=-3$$

A second-order system requires two initial conditions to choose from $h[0]$ and $h[1]$

2. Solution of the unit impulse response

[Ex.] The difference equation for a causal discrete LTI system is

$$y[k]+3y[k-1]+2y[k-2]=x[k]$$

Find the unit impulse response of the system $h[k]$

Solution: $h[k]$ satisfies the equation $h[k]+3h[k-1]+2h[k-2]=\delta[k]$

(3) Determine the pending coefficients for homogeneous solutions

$$h[k]=C_1(-1)^k+C_2(-2)^k, k\geq 0$$

Substitute the initial condition $h[0]=1 \quad h[1]=-3$

$$\begin{aligned} h[0]&=C_1+C_2=1, \\ h[1]&=-C_1-2C_2=-3 \end{aligned} \Rightarrow C_1=-1,\ C_2=2$$

$$h[k]=[-(-1)^k+2(-2)^k]u[k]$$

Zero-state response of discrete LTI systems

- ※ The system responds with zero state
- ※ Discrete convolution and computation
- ※ Discrete convolution and properties

1. System zero-state response

If the input signal is $x[k]$, the unit impulse response of the discrete LTI system is $h[k]$, Then the zero-state response of the system $y_{zs}[k]$ is:

$$y_{zs}[k] = x[k] * h[k] = \sum_{n=-\infty}^{\infty} x[n]h[k-n] \qquad \text{(Convolutional sum)}$$

It can be seen that the zero-state response of the discrete-time LTI system is the convolutional sum of the input signal and the system unit impulse response, which reveals the interaction mechanism between the signal and the system in the time domain.

2. Calculation of convolutional sums

Parsing method: Evaluate directly from the expression of the convolutional sum

$$x[k]*h[k]=\sum_{n=-\infty}^{\infty} x[n]h[k-n]$$

If the signals x[k] and h[k] can be expressed as analytic functions, the convolutional sum can be calculated using the analytic method.

2. Calculation of convolutional sums

[Ex.] Calculate $x[k]=\alpha^k u[k]$ and $h[k]=\beta^k u[k]$ convolutional sum

solution: $\alpha^k u[k] * \beta^k u[k]$

$$=\sum_{n=-\infty}^{+\infty} \alpha^n u[n]\cdot \beta^{k-n} u[k-n]$$

$$=\begin{cases} \sum_{n=0}^{k} \alpha^n \cdot \beta^{k-n} & k\geq 0 \\ 0 & k<0 \end{cases} \quad =\begin{cases} \dfrac{\beta^{k+1}-\alpha^{k+1}}{\beta-\alpha}u[k] & \alpha\neq\beta \\ (k+1)a^k u[k] & \alpha=\beta \end{cases}$$

2. Calculation of convolutional sums

Graphical method: $$x[k]*h[k]=\sum_{n=-\infty}^{\infty} x[n]h[k-n]$$

The steps to calculate convolutional sums graphically:

(1) Change the arguments in $x[k]$ and $h[k]$ from k to n;

(2) Flip one of the signals, such as flipping $h[n]$ to $h[-n]$;

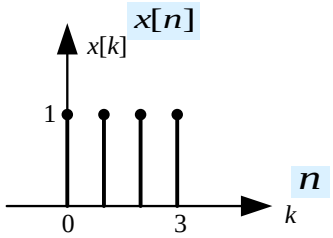
(3) Shift $h[-n]$ to k, where k is the parameter variable. The $k>0$ graph moves to the right and the $k<0$ graph moves to the left。

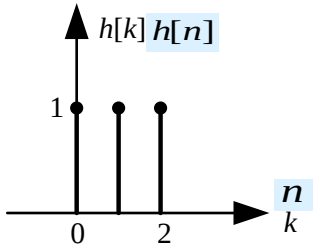
(4) Multiply $x[n]$ by $h[k-n]$.;

(5) Sum the graphs after the product。

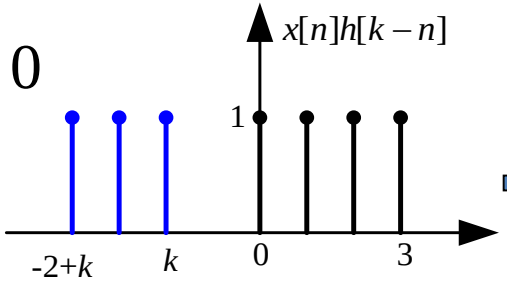
[Ex.] Calculate $y[k] = x[k] * h[k]$

Solution:

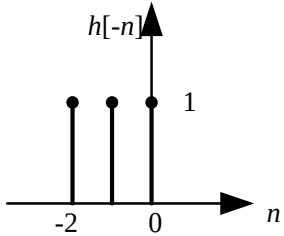




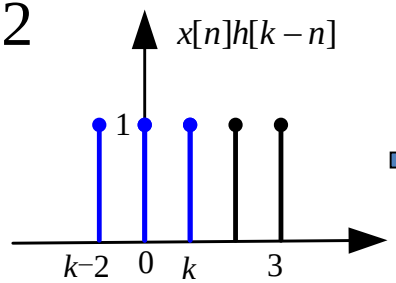
(a) $k < 0$



$$\Rightarrow y[k] = 0$$



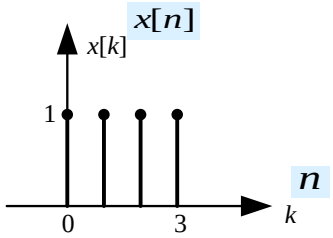
(b) $0 \le k < 2$

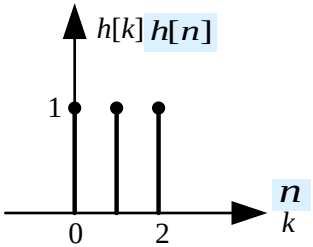


$$\Rightarrow y[k] = \sum_{n=0}^{k} 1 = k + 1$$

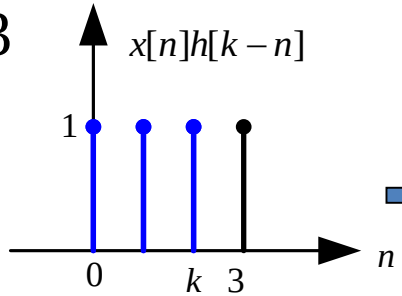
[Ex.] Calculate $y[k] = x[k] * h[k]$

Solution:

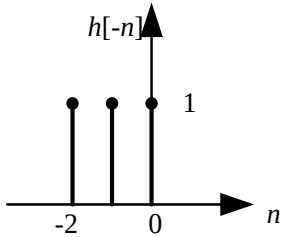




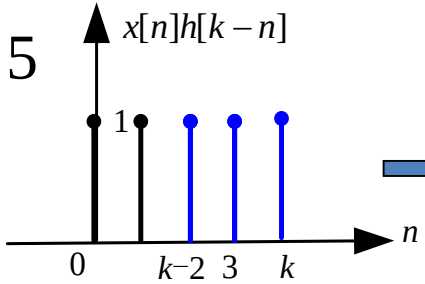
(c) $2 \le k \le 3$



$$\Rightarrow y[k] = \sum_{n=k-2}^{k} 1 = 3$$



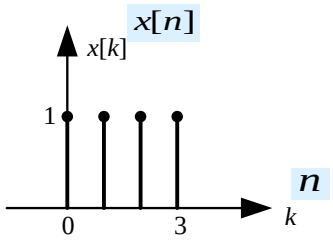
(d) $3 < k \le 5$

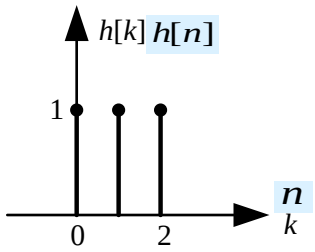


$$\Rightarrow y[k] = \sum_{n=k-2}^{3} 1 = 6 - k$$

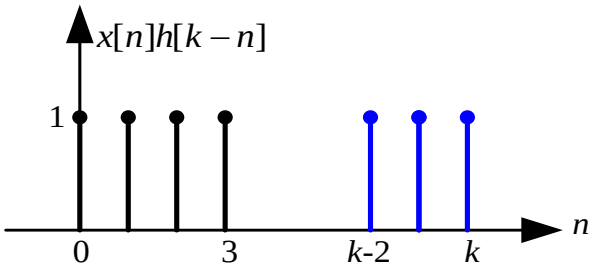
[Ex.] Calculate $y[k] = x[k] * h[k]$

Solution:

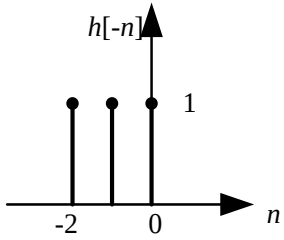




(e) $5 < k$

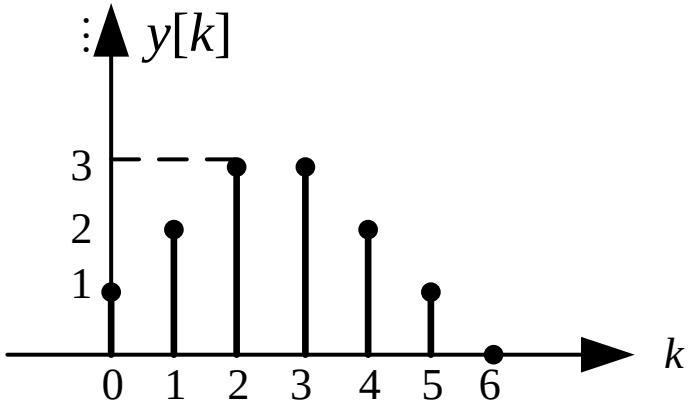


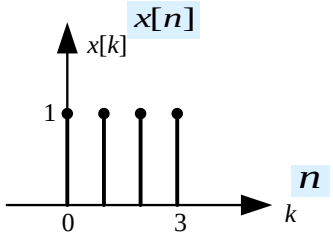
$\Rightarrow y[k] = 0$

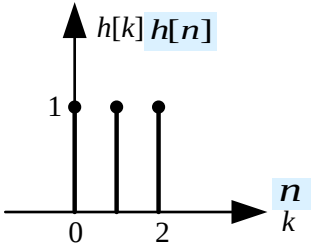


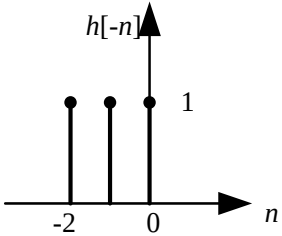
[Ex.] Calculate $y[k] = x[k] * h[k]$。

Solution：









The convolutional sum of the two signals, convolution and the result are still one signal. The starting point of the signal is equal to the sum of the beginnings of the two signals, and the end point is equal to the sum of the end points of the two signals.

2. Calculation of convolutional sums

List method: If $x[k]$ and $h[k]$ are both causal sequences, there are

$$y[k] = x[k] * h[k] = \sum_{n=0}^{k} x[n]h[k-n], k \geq 0$$

- When $k = 0$, $y[0] = x[0]h[0]$
- When $k = 1$, $y[1] = x[0]h[1] + x[1]h[0]$
- When $k = 2$, $y[2] = x[0]h[2] + x[1]h[1] + x[2]h[0]$
- When $k = 3$, $y[3] = x[0]h[3] + x[1]h[2] + x[2]h[1] + x[3]h[0]$

The above solution process can be summarized into a list method

2. Calculation of convolutional sums

The values of $h[k]$ are arranged in a row, and the values of $x[k]$ are arranged in a column

	$h[0]$	$h[1]$	$h[2]$	$h[3]$	$\cdots$
$x[0]$	$x[0]\,h[0]$	$x[0]\,h[1]$	$x[0]\,h[2]$	$x[0]\,h[3]$	$\cdots$
$x[1]$	$x[1]\,h[0]$	$x[1]\,h[1]$	$x[1]\,h[2]$	$x[1]\,h[3]$	$\cdots$
$x[2]$	$x[2]\,h[0]$	$x[2]\,h[1]$	$x[2]\,h[2]$	$x[2]\,h[3]$	$\cdots$
$x[3]$	$x[3]\,h[0]$	$x[3]\,h[1]$	$x[3]\,h[2]$	$x[3]\,h[3]$	$\cdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\cdots$

Each value on the diagonal slash is the value of $x[n]h[k-n]$

The sum of the values on the diagonal slash is the value of the $y[k]$ item。

2. Calculation of convolutional sums

[Ex.] Calculate $x[k]=\{1,\ 2,\ \overset{\downarrow}{0},\ 3,\ 2\}$ and $h[k]=\{1,\ \overset{\downarrow}{4},\ 2,\ 3\}$ convolution

solution:

		$x[-2]$	$x[-1]$	$x[0]$	$x[1]$	$x[2]$
		1	2	0	3	2
$h[-1]$	1	1	2	0	3	2
$h[0]$	4	4	8	0	12	8
$h[1]$	2	2	4	0	6	4
$h[2]$	3	3	6	0	9	6

$$y[k]=\{1,\ 6,\ 10,\ \overset{\downarrow}{10},\ 20,\ 14,\ 13,\ 6\}$$

The starting point of the convolutional sum is equal to the sum of the starting points of the two convolutional sequences, determining the starting point of the sequence。

3. The nature of convolutional sums

(1) Commutative law: $x[k] * h[k] = h[k] * x[k]$

(2) Binding law: $x[k] * \{ h_1[k] * h_2[k]\} = \{ x[k] * h_1[k] \} * h_2[k]$

(3) distributive law:

$$x[k] * \{ h_1[k] + h_2[k] \} = x[k] * h_1[k] + x[k] * h_2[k]$$

3. The nature of convolutional sums

(4) Displacement characteristics: If $x[k]*h[k]=y[k]$, then

$$x[k-n]*h[k-l]=y[k-(n+l)]$$

$$x[k]*\delta[k-n]=x[k-n]$$

(5) Difference and summation characteristics: If $x[k]*h[k]=y[k]$

$$\nabla x[k]*h[k]=x[k]*\nabla h[k]=\nabla y[k]$$

$$x[k]*\sum_{n=-\infty}^{k}h[n]=(\sum_{n=-\infty}^{k}x[n])*h[k]=\sum_{n=-\infty}^{k}y[n]$$

3. The nature of convolutional sums

[Ex.] The convolutional sum of $h[k]=\{1, \underset{\downarrow}{4}, 5, 3\}$ and $x[k]=\{1, 0, \underset{\downarrow}{2}, 4\}$ is calculated using the properties of convolution

solution: $$x[k]=\delta[k+2]+2\delta[k]+4\delta[k-1]$$

Take advantage of the distribution law and displacement characteristics

$$x[k]*h[k]=\{\delta[k+2]+2\delta[k]+4\delta[k-1]\}*h[k]$$

$$=h[k+2]+2h[k]+4h[k-1]$$

$$y[k]=x[k]*h[k]=\{1, 4, 7, \underset{\downarrow}{15}, 26, 26, 12\}$$

Thanks

Some of the materials cited in this course are the teaching accumulation of the main lecturer for many years, which come from a variety of media and the exchanges of colleagues, peers and friends, and it is difficult to indicate the source one by one, so I hereby explain and thank you!